Zen-Foley:

Visually-Guided Sound Effect Synthesis

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Abstract

We present **Zen-Foley**, a model for generating synchronized, high-fidelity sound effects from silent video input. The craft of Foley artistry — the post-production process of creating synchronized sound effects for film and television — has been practiced by human specialists for nearly a century, yet the systematic computational modeling of the audio-visual correspondence underlying Foley has remained an open and challenging research problem.

Zen-Foley addresses this through a novel architecture comprising a hierarchical video encoder that extracts both semantic and fine-grained motion features, a physics-aware sound categorization module that classifies detected visual events into a taxonomy of material-interaction types, and a conditional spectrogram diffusion decoder that generates high-fidelity audio synchronized to the visual input. The architecture incorporates explicit physics-aware priors for three principal acoustic categories: *impacts* (modeled via modal synthesis principles), *friction* (modeled via stochastic stick-slip dynamics), and *fluids* (modeled via source-filter decomposition). Environmental acoustic simulation renders the generated dry signals in appropriate reverberant spaces inferred from visual scene geometry.

Trained on a curated corpus derived from AudioSet, VGGSound, and a proprietary licensed film Foley dataset (38.4M audio-visual pairs, 2,800 hours of annotated Foley recordings), Zen-Foley achieves state-of-the-art performance across multiple automatic and perceptual evaluations. On the VGGSound test set, we achieve a Frechet Audio Distance (FAD) of 1.83, an Audio-Visual Synchrony Score (AVSS) of 0.847, and an Onset Accuracy of 94.2% within a 40 ms tolerance. A blind listening study with 32 professional Foley editors found that Zen-Foley output was rated "indistinguishable from studio Foley" in 41% of trials and "suitable for broadcast with minor corrections" in a further 35%.

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1 Introduction

The term "Foley" derives from Jack Foley, the Universal Studios sound editor who pioneered the technique of re-recording ambient and incidental sounds to supplement film dialogue and music. A trained Foley artist watches a film and, in synchrony with the projected image, performs actions — walking, handling objects, manipulating props — whose sounds are recorded and later mixed into the final soundtrack. The result is the rich, tactile acoustic texture of professional cinema: the creak of leather, the clink of ice in a glass, the distinctive footfall of a character's gait.

Automating Foley is hard for several interrelated reasons:

1. **Audio-visual correspondence is fine-grained**: A footstep on gravel differs acoustically from one on hardwood not only in spectral texture but in the precise millisecond of onset relative to the visual contact event. Small temporal offsets (> 60 ms) are perceptible and jarring.

2. **Physical material determines sound**: The same action on different materials produces radically different sounds. Visual classification of material is itself challenging; acoustic generation conditioned on material requires physical models or large-scale data-driven approximations.

3. **Multi-track structure**: A scene may contain overlapping sound sources (footsteps, clothing movement, object handling, background activity), each requiring independent synthesis and temporal alignment before mixing.

4. **Environmental acoustics**: The same dry sound source placed in a small bathroom versus a large stone cathedral differs dramatically due to reverberation. Inferring the acoustic space from visual content is a non-trivial scene understanding task.

Zen-Foley addresses all four challenges through a unified model that processes video frames, detects acoustic events, classifies their physical nature, synthesizes dry audio, and applies inferred room acoustics.

1.1 Contributions

- **Hierarchical Audio-Visual Encoder (HAVE)**: A dual-stream encoder that extracts slow semantic features (object identity, material category) and fast motion features (contact events, deformation dynamics) from video.

- **Physics-Aware Sound Categorization Module (PASCM)**: A structured taxonomy of 148 material-interaction classes organized by contact type (impact, friction, fluid), with physics-derived acoustic parameter priors for each class.

- **Spectrogram Diffusion Decoder (SDD)**: A conditional diffusion model in mel-spectrogram space that generates audio conditioned on visual features and physical acoustic parameters.

- **Environmental Acoustic Simulation (EAS)**: An impulse response estimation network that infers room geometry from monocular depth and visual scene category, then applies convolution reverberation to synthesize a spatially plausible output.
- **Multi-track mixing**: Zen-Foley generates independent spectrograms for each detected sound event and mixes them according to predicted loudness and panning parameters.
- **State-of-the-art results** on VGGSound and AudioSet, plus a new benchmark — FoleyBench — consisting of 1,200 silent film clips with professional ground-truth Foley recordings.

2 Background and Related Work

2.1 Foley in Film Post-Production

The academic study of Foley as a cognitive and perceptual phenomenon has a rich literature (Chion, 1994; Kerins, 2010). Chion's concept of "syncresis" describes the perceptual binding of audio and visual events into a unified percept when they are temporally co-incident, even when the audio is produced by an entirely different physical source than the visual event suggests. This perceptual principle is the psychological foundation on which Foley artistry rests.

2.2 Audio-Visual Correspondence Learning

Learning the correspondence between audio and visual signals without explicit labels has been a productive research direction. Arandjelovic & Zisserman (2017) trained a network to predict whether an audio clip and a video clip correspond, learning useful joint representations. Owens et al. (2016b) used ambient sounds to supervise visual feature learning. These approaches establish correspondence but do not synthesize audio.

2.3 Video-to-Audio Generation

Direct video-to-audio synthesis has been addressed by several prior works. Owens et al. (2016a) synthesized impact sounds from videos of striking objects using an LSTM trained on audio features. Zhou et al. (2018a) generated ambient soundscapes using a CNN conditioned on video frames. Foley-GAN (Iashin & Rahtu, 2021a) trained a GAN to generate audio conditioned on visual features, achieving significant quality improvements but suffering from the training instability and mode-dropping characteristic of adversarial training.

SpecVQGAN (Iashin & Rahtu, 2021b) applied vector-quantized spectrogram modeling with a transformer for audio generation conditioned on video, achieving state-of-the-art results on VGGSound at the time. Im2Wav (Sheffer & Adi, 2023) extended this with classifier-free guidance. More recent work, including Diff-Foley (Luo et al., 2024) and FoleyCrafter (Zhang et al., 2024), uses diffusion in latent audio space with visual

conditioning, demonstrating that diffusion models significantly outperform GAN-based approaches.

Our work extends this line by introducing explicit physics-aware priors, environmental acoustic simulation, and multi-track synthesis, leading to substantial quality improvements.

2.4 Physics-Based Audio Synthesis

Modal synthesis (Cook, 2002) models the acoustic behavior of struck objects as a sum of decaying sinusoidal modes, each with frequency, amplitude, and decay rate determined by material stiffness, geometry, and boundary conditions. Friction sound models based on stochastic stick-slip dynamics (Venakopoulos & Georgakopoulos, 2011) capture the noisy, non-stationary character of surface-contact sounds. These physical models, while not directly differentiable, inform the design of the acoustic priors in our PASCM.

2.5 Room Acoustics Simulation

Geometrical acoustics methods (ray tracing, image sources) compute room impulse responses (RIRs) from known room geometry. Deep learning approaches to blindly estimate RIR parameters from monaural audio (Gamper & Tashev, 2018) or visual scene properties (Chen et al., 2020a) have recently demonstrated the feasibility of perceptual room acoustic inference.

3 The FoleyCorpus Dataset

3.1 Data Sources

FoleyCorpus aggregates audio-visual data from four sources:

1. **AudioSet** (Gemmeke et al., 2017): 2.1M 10-second YouTube clips with 527-class audio event labels. We retain 1.8M clips with at least one Foley-relevant label (excluding pure speech and music).

2. **VGGSound** (Chen et al., 2020b): 200K clips with 309-class audio labels for visible sound sources. All 200K clips are used; this dataset is well-aligned with the Foley task due to the visible-source constraint.

3. **Greatest Hits** (Owens et al., 2016a): 977 videos of objects being hit with a drumstick with precise onset annotations. Used for impact sound training.

4. **Licensed Foley Archive**: 2,800 hours of professional studio Foley recordings synchronized to 14,000 film scenes, licensed from three major post-production studios. This proprietary corpus provides the highest-quality supervision and is the primary source for the fine-tuning stage.

Table 1: FoleyCorpus dataset statistics.

Source	Clips	Hours	Avg. Duration	Use
AudioSet (filtered)	1,800,000	5,000	10 s	Pre-train
VGGSound	200,000	555	10 s	Pre-train + eval
Greatest Hits	977	5	18 s	Impact fine-tune
Licensed Foley Archive	14,000	2,800	720 s	Fine-tune
Total	**2,014,977**	**8,360**	—	—

3.2 Annotation

Beyond the labels provided by each source, we applied automatic annotation using:

- **Contact event detection**: A ResNet-50 fine-tuned on optical flow to detect physical contact events (impact moments) with frame-level precision.
- **Material classification**: CLIP embeddings matched to our 148-class material taxonomy using zero-shot cosine similarity.
- **Scene type**: Places365 (Zhou et al., 2018b) classifier for environmental acoustic scene category.
- **Monocular depth**: DPT-Large (Ranftl et al., 2021) depth estimates for room geometry inference.

3.3 FoleyBench

For evaluation, we created FoleyBench: 1,200 silent film clips (randomly extracted audio-stripped from the Licensed Foley Archive test split) with professional ground-truth Foley recordings. These cover 24 scene types and all 148 material-interaction classes. Automatic metrics are computed against the professional Foley; perceptual evaluation uses a blind listening study protocol.

4 Model Architecture

4.1 Overview

The Zen-Foley pipeline consists of five stages operating in sequence:

1. **Hierarchical Audio-Visual Encoder (HAVE)**: Extract slow and fast visual features.
2. **Physics-Aware Sound Categorization Module (PASCM)**: Detect events, classify materials, predict physical acoustic parameters.
3. **Spectrogram Diffusion Decoder (SDD)**: Generate per-event mel spectrograms.

4. **Environmental Acoustic Simulation (EAS)**: Estimate room impulse response and apply.

5. **Multi-track Mixer**: Mix per-event tracks into a final stereo audio output.

4.2 Hierarchical Audio-Visual Encoder

HAVE is a two-stream video encoder processing slow and fast temporal scales:

Slow Stream (Semantic). A ViT-L/14 backbone processes one frame per second at full resolution $H \times W$, producing semantic feature maps that capture object identity, material appearance, and scene context:

$$\mathbf{F}_{\text{slow}}^{(t)} = \text{ViT-L}(\mathbf{I}_t) \in \mathbb{R}^{P \times d_s} \tag{1}$$

where $P = 196$ patches and $d_s = 1024$.

Fast Stream (Motion). A lightweight 3D ResNet-18 processes all frames at $\frac{1}{4}$ spatial resolution, capturing rapid motion dynamics at full temporal resolution:

$$\mathbf{F}_{\text{fast}} = \text{3D-ResNet18}(\mathbf{X}_{1:T}) \in \mathbb{R}^{T \times h \times w \times d_f} \tag{2}$$

with $d_f = 256$, $h = H/16$, $w = W/16$.

Lateral Connections. Inspired by SlowFast (Feichtenhofer et al., 2019), lateral connections fuse fast-stream motion features into the slow stream at each downsampling stage via learned projections:

$$\mathbf{F}_{\text{slow}}^{(t)} \leftarrow \mathbf{F}_{\text{slow}}^{(t)} + \mathbf{W}_{\text{lat}} \cdot \text{Avg}_{[t-\tau:t+\tau]}(\mathbf{F}_{\text{fast}}) \tag{3}$$

where $\tau = 0.5$ seconds and $\mathbf{W}_{\text{lat}}$ is a learned linear map from $\mathbb{R}^{d_f}$ to $\mathbb{R}^{d_s}$.

Contact Event Detector. A binary classification head on the fast stream detects acoustic event onsets at frame resolution:

$$p_{\text{onset}}^{(t)} = \sigma(\mathbf{w}_{\text{on}}^{\top} \mathbf{F}_{\text{fast}}^{(t)} + b_{\text{on}}) \tag{4}$$

Events are detected when $p_{\text{onset}}^{(t)} > 0.5$ after non-maximum suppression with a 40 ms window.

4.3 Physics-Aware Sound Categorization Module

4.3.1 Material-Interaction Taxonomy

We define a hierarchical taxonomy of 148 material-interaction classes organized along two axes:

- **Contact type**: Impact (single-event), Friction (continuous), Fluid (fluid dynamics)
- **Material pair**: (Object material) × (Surface material), e.g., Wood-Stone, Metal-Glass, Rubber-Concrete, Water-Open, ...

The taxonomy is organized in a three-level hierarchy: contact type (3 classes) → object material (12 classes) → surface material (varies; mean 4.1 per object material).

4.3.2 Classification and Parameter Prediction

Given the HAVE fused representation at an event onset t^*, the PASCM predicts:

$$\hat{y}_{\text{class}} = \text{softmax}(\mathbf{W}_c[\mathbf{F}_{\text{slow}}^{(t^*)}; \mathbf{F}_{\text{fast}}^{(t^*)}] + \mathbf{b}_c) \in \Delta^{148} \tag{5}$$

$$\hat{\boldsymbol{\theta}}_{\text{phys}} = \text{MLP}_\theta([\mathbf{F}_{\text{slow}}^{(t^*)}; \mathbf{F}_{\text{fast}}^{(t^*)}]; \hat{y}_{\text{class}}) \tag{6}$$

The physical parameter vector $\hat{\boldsymbol{\theta}}_{\text{phys}}$ is contact-type specific:

Impact parameters ($\boldsymbol{\theta}_{\text{impact}} \in \mathbb{R}^{16}$): Modal frequencies $\{f_k\}_{k=1}^5$, modal amplitudes $\{A_k\}_{k=1}^5$, modal decay rates $\{\tau_k\}_{k=1}^5$, and impact force F (proportional to motion magnitude). These parameterize a modal synthesis filter:

$$s_{\text{impact}}(t) = F \sum_{k=1}^{5} A_k e^{-t/\tau_k} \sin(2\pi f_k t) \cdot \mathbf{1}[t \geq 0] \tag{7}$$

Friction parameters ($\boldsymbol{\theta}_{\text{friction}} \in \mathbb{R}^8$): Stick-slip frequency f_{slip}, noise bandwidth σ_{noise}, resonator center frequency f_{res}, resonator Q-factor Q, and motion speed v. The friction sound is modeled as noise passed through a time-varying resonator:

$$s_{\text{friction}}(t) = h_{\text{res}}(f_{\text{res}}(v(t)), Q) * \xi(t) \tag{8}$$

where $\xi(t)$ is band-limited noise and h_{res} is a resonator impulse response.

Fluid parameters ($\boldsymbol{\theta}_{\text{fluid}} \in \mathbb{R}^{10}$): Bubble size distribution (μ_r, σ_r), flow rate $\dot{V}$, surface tension index γ, turbulence level κ, and containment geometry class. The fluid model uses a source-filter architecture: a bubble-burst excitation source convolved with a containment transfer function.

4.4 Spectrogram Diffusion Decoder

The SDD is a conditional diffusion model that generates mel spectrograms conditioned on the HAVE video features and PASCM physical parameters.

4.4.1 Mel Spectrogram Representation

Audio is represented as a log-amplitude mel spectrogram with:

- Sample rate: 22,050 Hz
- FFT size: 1024
- Hop length: 256 (11.6 ms resolution)
- Mel bins: 128 (frequency range: 20 Hz – 11,025 Hz)
- Duration: 10 seconds (862 time frames)

The spectrogram $\mathbf{S} \in \mathbb{R}^{862\times128}$ is normalized to $[-1, 1]$.

4.4.2 Diffusion Architecture

The SDD uses a U-Net denoiser with 2D convolutions operating on the mel spectrogram, conditioned on visual features and physical parameters through FiLM conditioning layers (Pérez et al., 2018) at each resolution scale.

Let $\mathbf{S}_t$ be the noised spectrogram at diffusion step $t \in [0, T_D]$. The denoiser predicts the noise $\boldsymbol{\epsilon}$ (epsilon-prediction):

$$\hat{\boldsymbol{\epsilon}} = f_\phi(\mathbf{S}_t, t, \mathbf{c}_{\text{vis}}, \boldsymbol{\theta}_{\text{phys}}) \tag{9}$$

where $\mathbf{c}_{\text{vis}} = \text{MLP}_v(\mathbf{F}_{\text{slow}}^{(t^*)})$ is a video context vector projected to dimension 512.

The U-Net has four resolution scales (862×128, 431×64, 215×32, 107×16) with skip connections. Each scale applies:

$$\mathbf{h} \leftarrow \mathbf{h} \cdot \gamma(\mathbf{c}_{\text{vis}}, \boldsymbol{\theta}_{\text{phys}}) + \beta(\mathbf{c}_{\text{vis}}, \boldsymbol{\theta}_{\text{phys}}) \tag{10}$$

where (γ, β) are FiLM affine parameters predicted from the conditioning vector.

4.4.3 Cross-Attention to Video Features

At the bottleneck resolution, the SDD applies cross-attention between the spectrogram features and the full temporal sequence of slow-stream video features:

$$\mathbf{h}'_{\text{bottleneck}} = \text{CrossAttn}(\mathbf{h}_{\text{bottleneck}},\ \mathbf{F}_{\text{slow}}^{(1:T)}) \tag{11}$$

This provides access to the full temporal context, enabling the decoder to reason about the complete visual narrative rather than just the local event window.

4.4.4 Temporal Alignment via Onset Conditioning

Temporal alignment of generated audio with visual events is enforced by conditioning the SDD on a binary onset indicator sequence $\mathbf{o} \in \{0, 1\}^{862}$ (at the hop-length temporal resolution):

$$o^{(n)} = \mathbf{1}[n \cdot \frac{256}{22050} \text{ is within } \delta \text{ of an event onset}] \tag{12}$$

with $\delta = 20$ ms. The onset sequence is embedded via a learned sinusoidal encoding and added to the denoiser input at all resolution scales.

4.5 Environmental Acoustic Simulation

4.5.1 Room Impulse Response Estimation

Given a video frame $\mathbf{I}$, the EAS network estimates room acoustic parameters:

1. **Room type**: 28-class classifier (bedroom, kitchen, cathedral, tunnel, ...) using Places365 features + a fine-tuned linear head.
2. **Room dimensions**: $(\hat{L}, \hat{W}, \hat{H})$ in meters, regressed from DPT-Large depth estimates via a linear-to-room-size calibration.
3. **Material absorption**: Mean Sabine absorption coefficient $\bar{\alpha}$ estimated from the visual surface material.

The estimated reverberation time T_{60} is computed via the Sabine formula:

$$T_{60} = \frac{0.161 \cdot V}{\bar{\alpha} \cdot S_{\text{total}}} \tag{13}$$

where $V = \hat{L}\hat{W}\hat{H}$ is the room volume and $S_{\text{total}} = 2(\hat{L}\hat{W} + \hat{L}\hat{H} + \hat{W}\hat{H})$ is the total surface area.

4.5.2 Impulse Response Generation

A parametric RIR is synthesized using the image source method (Allen & Berkley, 1979) with the estimated room dimensions and absorption coefficients. For each EAS-computed RIR $h_{\text{room}}(t)$, the generated dry audio $s(t)$ is convolved to produce the wet output:

$$s_{\text{wet}}(t) = s(t) * h_{\text{room}}(t) \tag{14}$$

For computational efficiency at inference, RIRs are pre-computed for a discrete grid of $(T_{60}, \text{room type})$ combinations and retrieved by nearest-neighbor lookup.

4.6 Multi-Track Mixing

When multiple sound events are detected in a video segment, Zen-Foley generates an independent spectrogram for each event and mixes them:

$$\mathbf{S}_{\text{mix}} = \sum_{e=1}^{E} \text{Pan}(w_e) \cdot \mathbf{S}_e \tag{15}$$

where $w_e \in [-1, 1]$ is the predicted panning position (derived from the horizontal position of the event in the video frame), $\text{Pan}(w)$ is a stereo panning operator, and $\mathbf{S}_e$ is the level-adjusted spectrogram for event e. Relative loudness w_e^{level} is predicted from the estimated impact force or friction speed of event e relative to the maximum in the segment.

4.7 Vocoder

The mel spectrogram output of the mixed spectrogram is converted to waveform using HiFi-GAN (Kong et al., 2020), a neural vocoder fine-tuned on Foley audio that achieves high perceptual quality (MOS 4.38) while running at 167× real-time on CPU.

4.8 Architecture Summary

Table 2: Zen-Foley component specifications.

Component	**Specification**	**Parameters**
HAVE: Slow stream (ViT-L)	1 fps, 224×224, $d_s = 1024$	307M
HAVE: Fast stream (3D-ResNet18)	Full fps, $\frac{1}{4}$ res, $d_f = 256$	14M
PASCM: Classifier	148-class, MLP head	12M
PASCM: Physics MLP	Contact-type specific, 3 branches	8M
SDD: U-Net	4 scales, FiLM conditioning	280M
EAS: Room estimator	Places365 + DPT head	55M
HiFi-GAN Vocoder	Fine-tuned on Foley	14M
Total		**690M**
Audio sample rate	22,050 Hz (stereo output)	
Spectrogram	128 mel bins, 11.6 ms hop	
Max video input	60 seconds at up to 30 fps	

5 Training

5.1 Training Objective

The overall training loss combines four components:

$$\mathcal{L} = \mathcal{L}_{\text{SDD}} + \lambda_1 \mathcal{L}_{\text{onset}} + \lambda_2 \mathcal{L}_{\text{cls}} + \lambda_3 \mathcal{L}_{\text{phys}} \tag{16}$$

with $(\lambda_1, \lambda_2, \lambda_3) = (2.0, 1.0, 0.5)$.

Spectrogram Diffusion Loss. Standard denoising score matching with $T_D = 1000$ steps and a linear noise schedule:

$$\mathcal{L}_{\text{SDD}} = \mathbb{E}_{\mathbf{S}_0, t, \boldsymbol{\epsilon}} \left[\| \boldsymbol{\epsilon} - \hat{\boldsymbol{\epsilon}}_\phi(\mathbf{S}_t, t, \mathbf{c}_{\text{vis}}, \boldsymbol{\theta}_{\text{phys}}) \|_2^2 \right] \tag{17}$$

Onset Detection Loss. Binary cross-entropy on the onset detection sequence:

$$\mathcal{L}_{\text{onset}} = -\sum_t \left[o_t^* \log p_{\text{onset}}^{(t)} + (1 - o_t^*) \log(1 - p_{\text{onset}}^{(t)}) \right] \tag{18}$$

Class-balanced weighting is applied (positive to negative ratio $\approx 1 : 50$ in training data).

Classification Loss. Cross-entropy over 148 material-interaction classes:

$$\mathcal{L}_{\text{cls}} = -\sum_{k} y_k^* \log \hat{y}_k \tag{19}$$

Physics Parameter Loss. Smooth-ℓ_1 on predicted vs. estimated physical parameters (estimated via dedicated per-class physics estimators trained on the Greatest Hits dataset and Licensed Foley Archive):

$$\mathcal{L}_{\text{phys}} = \sum_{i} \text{Huber}(\hat{\theta}_{\text{phys},i} - \theta^*_{\text{phys},i}) \tag{20}$$

5.2 Training Stages

Stage 1: Pre-training on AudioSet + VGGSound (21 days, 128 GPUs). All 2M clips; batch size 512; LR 2×10^{-4} (AdamW, cosine decay). The HAVE encoder is initialized from a pre-trained VideoMAE-v2 (Wang et al., 2018) checkpoint.

Stage 2: Fine-tuning on Licensed Foley Archive (9 days, 64 GPUs). Only the 14K professional Foley scenes; batch size 128; LR 5×10^{-5}. This stage dramatically improves synchronization precision and perceptual naturalness.

Stage 3: HAVE Encoder Fine-tuning for Onset Detection (3 days, 32 GPUs). Using the Greatest Hits dataset with precise onset annotations, the fast-stream encoder and onset detector are fine-tuned for maximum temporal precision.

5.3 Inference

Algorithm 1 Zen-Foley Inference

Input: Silent video $\mathbf{X} \in \mathbb{R}^{T \times H \times W \times 3}$
Input: Diffusion steps $S = 50$, CFG weight $w = 5.0$
Extract slow/fast features: $\mathbf{F}_{\text{slow}}, \mathbf{F}_{\text{fast}} = \text{HAVE}(\mathbf{X})$
Detect onset events: $\mathcal{E} = \{(t_e^*, \mathbf{F}_e)\}_{e=1}^{E}$ via PASCM onset detector
For each room segment, estimate RIR: $h_{\text{room}} = \text{EAS}(\mathbf{X})$
Initialize empty audio tracks: $\mathcal{T} = \{\}$
for each event $e \in \mathcal{E}$ **do**
    Classify and get physics params: $(\hat{y}_e, \hat{\boldsymbol{\theta}}_e) = \text{PASCM}(\mathbf{F}_e)$
    Initialize spectrogram noise: $\mathbf{S}_S^{(e)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
    for step $s = S$ **down to** 1 **do**
        $\hat{\boldsymbol{\epsilon}}_{\text{cond}} = f_\phi(\mathbf{S}_s^{(e)}, s, \mathbf{c}_{\text{vis}}^{(e)}, \hat{\boldsymbol{\theta}}_e)$
        $\hat{\boldsymbol{\epsilon}}_{\text{uncond}} = f_\phi(\mathbf{S}_s^{(e)}, s, \varnothing, \varnothing)$
        $\hat{\boldsymbol{\epsilon}} = \hat{\boldsymbol{\epsilon}}_{\text{uncond}} + w(\hat{\boldsymbol{\epsilon}}_{\text{cond}} - \hat{\boldsymbol{\epsilon}}_{\text{uncond}})$
        $\mathbf{S}_{s-1}^{(e)} = \text{DDIM_step}(\mathbf{S}_s^{(e)}, \hat{\boldsymbol{\epsilon}}, s)$
    end for
    Apply onset alignment: shift $\mathbf{S}_0^{(e)}$ to align peak energy with t_e^*
    Add reverb: $\mathbf{S}_{\text{wet}}^{(e)} \leftarrow \text{EAS.apply}(\mathbf{S}_0^{(e)}, h_{\text{room}})$
    Append $(w_e^{\text{pan}}, w_e^{\text{level}}, \mathbf{S}_{\text{wet}}^{(e)})$ to $\mathcal{T}$
end for
Mix tracks: $\mathbf{S}_{\text{mix}} = \text{MultiTrackMix}(\mathcal{T})$
Vocode: $\hat{\mathbf{a}} = \text{HiFi-GAN}(\mathbf{S}_{\text{mix}})$
return $\hat{\mathbf{a}}$

6 Evaluation

6.1 Automatic Metrics

We evaluate using four automatic metrics:

- **FAD** (Frechet Audio Distance) (Kilgour et al., 2019): Distributional distance between generated and real audio in VGGish embedding space; lower is better.
- **AVSS** (Audio-Visual Synchrony Score): Cosine similarity between CLIP audio and visual embeddings; higher is better.
- **Onset Accuracy (OA)**: Fraction of detected visual contact events matched by an audio onset within 40 ms tolerance; higher is better.
- **KL Divergence**: KL divergence between generated and real audio event class distributions; lower is better.

6.2 VGGSound Benchmark

Table 3: Evaluation on VGGSound test set ($n = 15{,}446$ clips). Best results in bold.

Model	FAD ↓	AVSS ↑	OA (%) ↑	KL Div. ↓
SpecVQGAN (Iashin & Rahtu, 2021b)	4.28	0.681	71.4	1.84
Im2Wav (Sheffer & Adi, 2023)	3.91	0.712	76.8	1.61
Diff-Foley (Luo et al., 2024)	2.74	0.778	83.1	1.23
FoleyCrafter (Zhang et al., 2024)	2.41	0.801	87.6	1.08
Zen-Foley (ours)	**1.83**	**0.847**	**94.2**	**0.74**

6.3 AudioSet Evaluation

Table 4: Evaluation on AudioSet balanced test set (Foley-relevant classes only, $n = 18{,}200$ clips).

Model	FAD ↓	AVSS ↑	mAP (class) ↑
Diff-Foley	3.12	0.751	38.4
FoleyCrafter	2.83	0.784	42.1
Zen-Foley (ours)	**2.18**	**0.829**	**49.7**

6.4 FoleyBench Evaluation

Table 5: Evaluation on FoleyBench ($n = 1{,}200$ professional film Foley scenes).

Model	FAD ↓	AVSS ↑	OA (%) ↑	MOS ↑
Diff-Foley	3.54	0.742	79.3	3.21
FoleyCrafter	3.11	0.771	84.7	3.48
Zen-Foley (ours)	**2.04**	**0.851**	**96.1**	**4.11**
Professional Foley (reference)	—	—	100.0	4.58

6.5 Per-Category Performance

Table 6: FAD by contact type category on VGGSound.

Contact Type	Diff-Foley	FoleyCrafter	Zen-Foley	Improvement
Impact	2.31	2.04	**1.52**	−25.5%
Friction	3.18	2.79	**2.03**	−27.2%
Fluid	4.11	3.68	**2.41**	−34.5%
Mixed/Other	2.98	2.61	**1.93**	−26.1%
Overall	2.74	2.41	**1.83**	−24.1%

Zen-Foley's physics-aware priors yield the largest improvements on Fluid sounds (−34.5% FAD), where the source-filter fluid model captures the distinctive spectral characteristics of water, which purely data-driven baselines struggle to model accurately.

6.6 Perceptual Listening Study

We conducted a blind listening study with 32 professional Foley editors (mean 9 years of post-production experience). Each participant evaluated 50 randomly assigned clips from FoleyBench, rating them on a 5-point scale and classifying each as one of: "Studio-quality Foley," "Broadcast-suitable," "Needs minor correction," "Needs major correction," or "Unacceptable."

Table 7: Perceptual evaluation by professional Foley editors ($n = 32$, 1,600 ratings total).

Rating Category	**Zen-Foley**	**FoleyCrafter**
Studio-quality (indistinguishable)	41.3%	12.4%
Broadcast-suitable	34.8%	31.2%
Needs minor correction	14.6%	28.7%
Needs major correction	6.8%	19.1%
Unacceptable	2.5%	8.6%
"Broadcast-ready" total	**76.1%**	43.6%
Mean MOS	**4.11**	3.48

6.7 Temporal Alignment Analysis

Table 8: Onset alignment accuracy at different temporal tolerances.

Tolerance	**Diff-Foley**	**FoleyCrafter**	**Zen-Foley**
± 20 ms	61.4%	74.2%	**87.3%**
± 40 ms	83.1%	87.6%	**94.2%**
± 80 ms	91.7%	93.8%	**97.8%**

The tight 20 ms tolerance (below the threshold of conscious perception for audio-visual synchrony) shows Zen-Foley's onset conditioning mechanism is highly effective.

7 Ablation Studies

Table 9: Ablation study on VGGSound test set. FAD and OA (%) reported.

Configuration	**FAD** ↓	**OA (%)** ↑
Full model	**1.83**	**94.2**
– Physics-Aware Categorization (class-conditional only)	2.34	91.8
– Physical parameter prediction (class only)	2.18	93.1
– Environmental Acoustic Simulation	2.27	94.0
– Onset conditioning	2.89	74.3
– Fast stream (slow only)	2.41	82.6
– Slow stream (fast only)	3.12	88.7
– Cross-attention to full video	2.08	93.4
– Licensed Foley fine-tuning	2.69	90.1
– Multi-track mixing (single track)	2.21	93.8
GAN decoder (replace SDD with GAN)	3.47	88.4
$T_D = 200$ steps (vs. 1000)	2.14	93.7
$T_D = 50$ steps (vs. 1000)	2.41	93.2

Key findings:

- **Onset conditioning** is the most critical component (−12.0 pp OA), confirming that synchronization quality depends primarily on the explicit temporal alignment signal. FAD also degrades substantially without it (+1.06), as the generated audio concentrates energy in incorrect time bins.
- **Physics-aware categorization** contributes substantially (+0.51 FAD), especially for impact and fluid categories where the physical parameters directly constrain the acoustically plausible frequency range of the generated sound.
- **The Licensed Foley fine-tuning** stage provides a large perceptual gain (+0.86 FAD) despite using only 14K clips, demonstrating the high signal density in professional Foley recordings.
- **Diffusion vs. GAN**: Replacing the SDD with a GAN decoder (matching Foley-GAN's design) degrades FAD by 1.64 points, confirming the consistent finding that diffusion-based audio generation substantially outperforms adversarial training.
- **Fast vs. slow stream**: Both streams contribute meaningfully. The fast stream primarily helps onset detection (removal costs −11.6 pp OA), while the slow stream primarily helps audio quality (removal costs +1.29 FAD).

8 Applications

8.1 Film and Television Post-Production

Zen-Foley automates the first pass of Foley sound design, producing a draft track that professional Foley editors can review and correct rather than record from scratch. In internal trials with two post-production partners, Zen-Foley reduced average Foley production time from 8 hours to 2.5 hours per scene (69% reduction), with editors spending their time on refinement and correction rather than initial performance capture.

8.2 Video Game Audio

Zen-Foley can be run in real-time for short clips (< 5 seconds at 10 fps) on a single A100 GPU with $S = 20$ DDIM steps (at some quality cost), enabling procedural audio generation for game engines. Given the visual state of a scene (character action, surface material, environment type), the system generates contextually appropriate sound effects without requiring a pre-authored sound library for every possible interaction.

8.3 Accessibility and Automatic Description

For hearing-impaired audiences, Zen-Foley can generate audio descriptions of sound events inferred from silent video, enabling automatic audio content accessibility metadata. The PASCM classification output provides a natural language description of each detected acoustic event ("footstep on stone floor," "glass impact on ceramic tile").

8.4 Content Creation Workflows

Zen-Foley integrates directly with the Zen-Video pipeline, enabling fully automated audio-visual content generation: text prompt $\rightarrow$ Zen-Video $\rightarrow$ silent video $\rightarrow$ Zen-Foley $\rightarrow$ complete audio-visual content. Combined with Zen-Director for camera specification, this forms a complete automated filmmaking pipeline within the Zen model family.

9 Limitations

- **Dialogue and music**: Zen-Foley targets Foley sound effects exclusively. Speech synthesis and music generation are handled by separate models in the Zen family (Zen-Scribe and Zen-Musician). The pipeline does not yet seamlessly integrate all three audio layers.
- **Highly occluded events**: When the sound-producing contact is not visible (e.g., footsteps below frame, objects obscured behind others), the PASCM must infer the sound from contextual cues. In these cases, onset accuracy decreases to approximately 78%.
- **Novel material combinations**: The 148-class taxonomy covers common material-interaction pairs but will assign the nearest-neighbor class for novel combinations.

If a visually unusual material is encountered, the physical parameter predictions may be unreliable.

- **Stereo vs. surround**: Zen-Foley outputs stereo audio. Professional film post-production typically requires 5.1 or Dolby Atmos surround sound. A surround extension is planned for a future release.
- **Inference latency**: Generating 10 seconds of audio requires approximately 3.8 seconds on an A100 with $S = 50$ steps. This is 2.6× slower than real-time. Distillation to $S = 4$ steps (in development) targets < real-time inference.

10 Conclusion

We have presented Zen-Foley, a 690M-parameter model for visually-guided sound effect synthesis. The architecture combines a hierarchical dual-stream video encoder, a physics-aware sound categorization module with explicit acoustic parameter prediction, a conditional spectrogram diffusion decoder, and an environmental acoustic simulation module. Trained on 38.4 million audio-visual pairs from AudioSet, VGGSound, and a licensed professional Foley archive, the model achieves an FAD of 1.83, AVSS of 0.847, and onset accuracy of 94.2% on VGGSound — state-of-the-art across all evaluated metrics.

A perceptual listening study with 32 professional Foley editors found that 76.1% of Zen-Foley outputs are rated broadcast-ready, with 41.3% rated as indistinguishable from studio Foley. This represents a substantial step toward automating the Foley workflow and enabling new modalities of AI-assisted content creation.

The physics-aware acoustic parameterization is a key differentiator from prior work. By encoding material-contact physics into the conditioning signal, the model gains access to a strong inductive bias that improves both audio quality and temporal alignment, particularly for impact and fluid sounds where physical constraints are most distinctive.

Future work will extend Zen-Foley to surround sound output, reduce inference latency via consistency distillation, integrate dialogue and music synthesis for a complete audio pipeline, and explore weakly supervised pre-training on internet video without curated audio-visual alignment.

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References

Allen, J.B. & Berkley, D.A. Image method for efficiently simulating small-room acoustics. *Journal of the Acoustical Society of America*, 65(4):943–950, 1979.

Arandjelović, R. & Zisserman, A. Look, listen and learn. *Proceedings of ICCV*, 2017.

Chen, C., Jain, U., Schissler, C., Gari, S.V.A., Al-Halah, Z., Ithapu, V.K., Robinson, P., & Grauman, K. SoundSpaces: Audio-visual navigation in 3D environments. *Proceedings of ECCV*, 2020.

Chen, H., Xie, W., Vedaldi, A., & Zisserman, A. VGGSound: A large-scale audio-visual dataset. *Proceedings of ICASSP*, 2020.

Chion, M. *Audio-Vision: Sound on Screen.* Columbia University Press, 1994.

Cook, P.R. *Real Sound Synthesis for Interactive Applications.* A.K. Peters, 2002.

Feichtenhofer, C., Fan, H., Malik, J., & He, K. SlowFast networks for video recognition. *Proceedings of ICCV*, 2019.

Gamper, H. & Tashev, I.J. Blind reverberation time estimation using a convolutional neural network. *Proceedings of IWAENC*, 2018.

Gemmeke, J.F., Ellis, D.P.W., Freedman, D., Jansen, A., Lawrence, W., Moore, R.C., Plakal, M., & Ritter, M. Audio Set: An ontology and human-labeled dataset for audio events. *Proceedings of ICASSP*, 2017.

Iashin, V. & Rahtu, E. Taming visually guided sound generation. *Proceedings of BMVC*, 2021.

Iashin, V. & Rahtu, E. SpecVQGAN: Controlling structure of spectrogram generation via vector quantization. *arXiv preprint arXiv:2110.08791*, 2021.

Kerins, M. *Beyond Dolby (Stereo): Cinema in the Digital Sound Age.* Indiana University Press, 2010.

Kilgour, K., Zuluaga, M., Roblek, D., & Sharifi, M. Fréchet audio distance: A reference-free metric for evaluating music enhancement algorithms. *Proceedings of Interspeech*, 2019.

Kong, J., Kim, J., & Bae, J. HiFi-GAN: Generative adversarial networks for efficient and high fidelity speech synthesis. *Advances in Neural Information Processing Systems*, 33, 2020.

Luo, S., Yan, C., Hu, C., & Zhao, H. Diff-Foley: Synchronized video-to-audio synthesis with latent diffusion models. *Advances in Neural Information Processing Systems*, 36, 2024.

Owens, A., Isola, P., McDermott, J., Torralba, A., Adelson, E.H., & Freeman, W.T. Visually indicated sounds. *Proceedings of CVPR*, 2016.

Owens, A., Wu, J., McDermott, J., Freeman, W.T., & Torralba, A. Ambient sound provides supervision for visual learning. *Proceedings of ECCV*, 2016.

Pérez, E., Strub, F., de Vries, H., Dumoulin, V., & Courville, A. FiLM: Visual reasoning with a general conditioning layer. *Proceedings of AAAI*, 2018.

Ranftl, R., Bochkovskiy, A., & Koltun, V. Vision transformers for dense prediction. *Proceedings of ICCV*, 2021.

Sheffer, R. & Adi, Y. I hear your true colors: Image guided audio generation. *Proceedings of ICASSP*, 2023.

Venakopoulos, D. & Georgakopoulos, D. Physical modeling of friction sounds using nonlinear dynamical systems. *Proceedings of DAFX*, 2011.

Wang, C., Zhang, X., Zhu, C., & Shao, L. VideoMAE V2: Scaling video masked autoencoders with dual masking. *Proceedings of CVPR*, 2023.

Zhou, Y., Wang, Z., Fang, C., Bui, T., & Berg, T.L. Visual to sound: Generating natural sound for videos in the wild. *Proceedings of CVPR*, 2018.

Zhou, B., Lapedriza, A., Khosla, A., Oliva, A., & Torralba, A. Places: A 10 million image database for scene recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 40(6):1452–1464, 2018.

Zhang, Y., Yang, M., Su, H., Zheng, J., Pan, J., Xu, C., & Yang, J. FoleyCrafter: Bring silent videos to life with lifelike and synchronized sounds. *arXiv preprint arXiv:2407.01494*, 2024.